

Kaushic Aravind B

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EXPERIENCE

ML Engineer | Scimplify

Jun 2025 - Present | Bangalore, IN

- Built **Scira**, a production **LangGraph research-agent ecosystem** powering **300 scientists** with domain-specific foundation models, planner delegation and MCP tools; reduced pre-wet-lab research cycles from **5 days to 1 hour** through a closed evaluation loop.
- Built the agent runtime with Landlock-sandboxed tools and row-level ACLs for multi-tenant isolation; added **LangSmith LLM-as-judge evaluation gates**.
- Built a search workflow using **SFT/LoRA-tuned Gemma 2 27B for structured output** (Unsloth) and **AWS OpenSearch hybrid BM25/vector retrieval**, metadata filters and cross-encoder reranking; **+30 recall points and 80% lower retrieval p90 latency** vs. vector-only search, with **model p50 latency cut from 15s to 3s** at 10K queries/day.
- Used a **Laya model for simpler NER** within the same search workflow.
- Designed agent memory using Semantica graph memory and a Letta-style file system with offline consolidation and **conflict reconciliation**.
- Built a chemical knowledge graph with **entity resolution, deduplication and write-time validation**; engineered a **PySpark ingestion pipeline loading ~1 TB into BigQuery in 2 hours**.
- Integrated **Redis LangCache** for semantic caching to skip inference on near-duplicate queries at **10K+ queries/day**.
- Trained a **Transformer + MCTS for chemical retrosynthesis**; achieved **88% top-1** and cut chemical-design cycles from **4 days to 15 minutes**.

Data Scientist | Sulekha

Sep 2023 - Jun 2025 | Chennai, IN

- Built QueryLM, a fine-tuned text-to-SQL system for business questions without schema exposure, shipped as a **cloud-hosted API**; cut **RAG cost 60%** through rechunking, caching and pipeline redesign.
- Built a hotel booking agent that answers queries and completes reservations through a payment gateway, cutting **inbound calls 80%**.
- Architected a Gemini audio platform processing **100K+ agent-hours/month**.
- Built a multi-task POS-BERT-BLSTM-CRF model for review opinion mining; automated categorization and routing cut **grievance resolution time 30%** and raised **repeat visits 40%**.

Applied AI Practitioner | Searce Inc

Jul 2022 - Sep 2023 | Bangalore, IN

- Built document-intelligence RAG on Pinecone with **retrieval and response-quality evals**; led solution architecture across **4+ enterprise GCP engagements**.
- Delivered an end-to-end **real-time speech recognition pipeline for ISRO**, the Indian Space Research Organisation, using Google STT with a live dashboard.
- Migrated **10+ years** of HubSpot candidate records to a structured analytics store with resume parsing and hybrid retrieval.

INDEPENDENT PROJECT

Local-first personal agent | Independent build

2026

- Designed and built a laptop-based agent with **Claude Agent SDK**, persistent git-tracked markdown memory, and a headed Chromium browser the owner can watch and take over.
- Kept logins in **macOS Keychain** so passwords are not exposed to the model; a deterministic, action-specific **one-time-code gate** protects sensitive actions.
- ~**4,550 lines of Python/HTML** and **95 offline tests** across memory, safety, vault and operations (project write-up).

TECHNICAL SKILLS

Agents & evals: LangGraph, LangChain, LangSmith, MCP, FastAPI, LLM-as-judge, benchmark design, A/B experimentation, agent memory, sandboxing

Retrieval & ML: OpenSearch, Pinecone, Milvus, pgvector, BM25 + vector, reranking, PyTorch, Hugging Face, SFT, LoRA, QLoRA, Unsloth

Data & engineering: Python, SQL, AWS, GCP, Docker, Redis LangCache, PostgreSQL, MongoDB, PySpark, BigQuery, knowledge graphs, entity resolution

EDUCATION & CERTIFICATIONS

B.Tech, Information Technology | College of Engineering Guindy, Anna University, Chennai | 2018 - 2022

Google Cloud Professional Machine Learning Engineer (2023) | Google Cloud Associate Cloud Engineer (2023)